

3.1. 이항분포 $B(n, p)$ 의 확률밀도함수

$$f(x) = \binom{n}{x} p^x (1-p)^{n-x}, \quad x = 0, 1, 2, \dots, n$$

에 대하여 부등식 $f(x) \geq f(x-1)$ 을 풀고, 이로부터 $f(x)$ 를 최대화하는 x 값을

$$x = \lfloor (n+1)p \rfloor : (n+1)p \text{ 이하의 정수}$$

임을 밝혀라. 이와 같이 확률밀도함수를 최대화 하는 값을 최빈값(mode)이라고 한다.

$$\begin{aligned} f(x) \geq f(x-1) &\Leftrightarrow \binom{n}{x} p^x (1-p)^{n-x} \geq \binom{n}{x-1} p^{x-1} (1-p)^{n-(x-1)} \\ &\Rightarrow \frac{n!}{x!(n-x)!} p^x (1-p)^{n-x} \geq \frac{n!}{(x-1)!(n-x+1)!} p^{x-1} (1-p)^{n-(x-1)} \\ &\Rightarrow (n-x+1)p \geq x(1-p) \\ &\Rightarrow np - xp + p \geq x(1-p) \\ &\Rightarrow np + p \geq x - xp + xp \\ &\Rightarrow p(n+1) \geq x \end{aligned}$$

3.2. 확률변수 X 가 이항분포 $B(n, p)$ 를 따를 때

$$E[X(X-1) \cdots (X-r+1)] = n(n-1) \cdots (n-r+1) p^r \quad (r = 1, 2, \dots, n)$$

임을 밝혀라.

$$\begin{aligned} E[X(X-1) \cdots (X-r+1)] &= \sum_{x=0}^{\infty} X(X-1) \cdots (X-r+1) \binom{n}{x} p^x (1-p)^{n-x} \\ &= \sum_{x=0}^{\infty} X(X-1) \cdots (X-r+1) \frac{n!}{x!(n-x)!} p^x (1-p)^{n-x} \\ &= \sum_{x=0}^{\infty} X(X-1) \cdots (X-r+1) \frac{n(n-1) \cdots (n-r+1)(n-r)!}{x!(n-x)!(n-r)!} p^x (1-p)^{n-x} \\ &= n(n-1) \cdots (n-r+1) p^r \sum_{x=r}^{\infty} \frac{(n-r)!}{(x-r)!(n-x)!} p^{x-r} (1-p)^{n-x} \\ \text{Let } y &= x-r \\ &= n(n-1) \cdots (n-r+1) p^r \sum_{y=0}^{\infty} \frac{(n-r)!}{y!(n-y-r)!} p^y (1-p)^{n-r-y} \\ \text{Let } m &= n-r \\ &= n(n-1) \cdots (n-r+1) p^r \sum_{y=0}^{\infty} \frac{m!}{y!(m-y)!} p^y (1-p)^{m-y} = (n)(n-1) \cdots (n-r+1) p^r \cdot 1 \end{aligned}$$

★★

3.3. 확률변수 X 가 이항분포 $B(n, p)$ ($0 < p < 1$)를 따를 때 연습문제 (1.19)에서 정의된 X 의 왜도와 첨예도를 구하라.

왜도: Skewness, 첨예도: Kurtosis

$$X \sim B(n, p), \quad f_X(x) = \binom{n}{x} p^x (1-p)^{n-x}$$

$$M_X(t) = M_{Z_1 + \dots + Z_n} = M_{Z_1}(t) \cdots M_{Z_n}(t), \quad Z_i \sim \text{Bernoulli}(p)$$

$$f_Z(z) = p^z (1-p)^{1-z}, \quad z = 0, 1$$

$$M_Z(t) = E[e^{tZ}] = \sum_{z \in \{0,1\}} e^{tZ} p^z (1-p)^{1-z} = e^t p + (1-p)$$

$$\therefore M_X(t) = (M_Z(t))^n = \{e^t p + q\}^n$$

$$G_X(t) = \log M_X(t) = n \log(e^t p + q)$$

$$G_X'(t) = n e^t p (e^t p + q)^{-1}$$

$$G_X''(t) = n e^t p (e^t p + q)^{-1} - n (e^t p)^2 (e^t p + q)^{-2}$$

$$G_X'''(t) = n e^t p (e^t p + q)^{-1} - n (e^t p)^2 (e^t p + q)^{-2} - 2n (e^t p)^2 (e^t p + q)^{-2} + 2n (e^t p)^3 (e^t p + q)^{-3}$$

$$\begin{aligned} G_X^{(4)}(t) &= n e^t p (e^t p + q)^{-1} - n (e^t p)^2 (e^t p + q)^{-2} \\ &\quad - 6n (e^t p)^2 (e^t p + q)^{-2} + 6n (e^t p)^3 (e^t p + q)^{-3} \\ &\quad + 6n (e^t p)^3 (e^t p + q)^{-3} - 6n (e^t p)^4 (e^t p + q)^{-4} \end{aligned}$$

$$G_X^{(1)}(t=0) = n p (p+q)^{-1} = n p (1-p)^{-1} = np - 3np^2 + 2np^3$$

$$\begin{aligned} G_X^{(2)}(t=0) &= n p (p+q)^{-1} - n p^2 (p+q)^{-2} = np(2p-3p+1) = np(2p-1)(p-1) \\ &\quad - 6np^2(p+q)^{-2} + 6np^3(p+q)^{-3} \\ &\quad + 6np^3(p+q)^{-3} - 6np^4(p+q)^{-4} \\ &= np - np^2 - 6np^2 + 6np^3 + 6np^3 - 6np^4 \\ &= np(1-7p+6p^2-6p^3) \end{aligned}$$

1.20.

표준화된 확률변수 $Z = (X-\mu)/\sigma$ 의 누물생성함수와 X 의 누물생성함수의 관계로부터 $Z = (X-\mu)/\sigma$ 의 누물과 X 의 누물 사이에 다음 관계가 성립하는 것을 밝혀라.

$$c_r\left(\frac{X-\mu}{\sigma}\right) = c_r(X)/\sigma^r \quad (r = 3, 4, \dots)$$

$$\therefore \text{왜도 (Skewness)} = C_3/\sigma^3 = C_3/[npq]^{3/2} \quad (\because \sigma^2 = npq)$$

$$\begin{aligned} &= np(2p-1)(p-1)/n \cdot n^{1/2} \cdot p \cdot p^{1/2} \cdot q \cdot q^{1/2} \\ &= 2p(2p-1) \cdot -q / \sqrt{npq} \\ &= (1-2p)/\sqrt{npq} \end{aligned}$$

$$\begin{aligned} \therefore \text{첨예도 (Kurtosis)} : C_4/\sigma^4 &= np(1-7p+6p^2-6p^3)/n^2 p^2 q^2 \\ &= (1-7p+6p^2-6p^3)/npq^2 \end{aligned}$$

3.4. $X = (X_1, \dots, X_k)^t \sim \text{Multi}(n, (p_1, \dots, p_k)^t)$ 일 때 다음을 밝히라.

(a) $(X_1, \dots, X_r, n - X_1 - \dots - X_r)^t \sim \text{Multi}(n, (p_1, \dots, p_r, 1 - p_1 - \dots - p_r)^t) \quad (1 \leq r \leq k)$

(b) $X_1 = x_1, \dots, X_r = x_r$ 인 조건에서 $(X_{r+1}, \dots, X_k)^t$ 의 조건분포를 유도하라

$$f_{X_{r+1}, \dots, X_k | X_1, \dots, X_r}(x_{r+1}, \dots, x_k | x_1, \dots, x_r)$$

를 구하라.

$$\begin{aligned} \text{(a)} \quad f_{X_1, \dots, X_k}(x_1, \dots, x_k) &= \binom{n}{x_1, \dots, x_k} p_1^{x_1} \dots p_k^{x_k} \\ f_{X_1, \dots, X_r, Z}(x_1, \dots, x_r, z) &= \binom{n}{x_1, \dots, x_r, z} p_1^{x_1} p_2^{x_2} \dots (1 - p_1 - \dots - p_r)^z \\ &= \binom{n}{x_1, \dots, x_r, z} p_1^{x_1} p_2^{x_2} \dots (1 - p_1 - \dots - p_r)^z \\ \therefore \text{pdf of } \text{Multi}(n, (p_1, p_2, \dots, (1 - p_1 - \dots - p_r))^t) \\ \text{by the uniqueness property of pdf,} \\ (X_1, \dots, X_r, n - X_1 - \dots - X_r) &\sim \text{Multi}(n, (p_1, p_2, \dots, (1 - p_1 - \dots - p_r))^t) \end{aligned}$$

[Textbook Solution]

$$\begin{aligned} \text{(a)} \quad (X_1, X_2, \dots, n - X_1 - \dots - X_r)^t &\stackrel{d}{=} \bigoplus_{i=1}^n \text{Multi}_k(1, (p_1, \dots, p_r, 1 - p_1 - \dots - p_r)^t) \\ &\stackrel{d}{=} \text{Multi}_k(n, (p_1, \dots, p_r, 1 - p_1 - \dots - p_r)^t) \end{aligned}$$

$$\begin{aligned} \text{(b)} \quad f_{X_{r+1}, \dots, X_k | X_1, \dots, X_r}(x_{r+1}, \dots, x_k | x_1, \dots, x_r) &= f_{X_1, \dots, X_r, X_{r+1}, \dots, X_k}(x_1, \dots, x_r, x_{r+1}, \dots, x_k) / f_{X_1, \dots, X_r}(x_1, \dots, x_r) \\ f_{X_1, \dots, X_r}(x_1, \dots, x_r) &= \binom{n}{x_1, \dots, x_r, n - x_1 - \dots - x_r} p_1^{x_1} p_2^{x_2} \dots p_r^{x_r} (1 - p_1 - \dots - p_r)^{n - x_1 - \dots - x_r} \\ f_{X_1, \dots, X_k}(x_1, \dots, x_k) &= \binom{n}{x_1, \dots, x_k} p_1^{x_1} \dots p_k^{x_k} \\ f_{X_{r+1}, \dots, X_k | X_1, \dots, X_r}(x_{r+1}, \dots, x_k | x_1, \dots, x_r) &= \frac{\binom{n}{x_1, \dots, x_r, x_{r+1}, \dots, x_k} p_1^{x_1} \dots p_k^{x_k}}{\binom{n}{x_1, \dots, x_r, n - x_1 - \dots - x_r} p_1^{x_1} p_2^{x_2} \dots p_r^{x_r} (1 - p_1 - \dots - p_r)^{n - x_1 - \dots - x_r}} \\ &= \frac{(n - x_1 - \dots - x_r)!}{x_{r+1}! \dots x_k!} \frac{p_{r+1}^{x_{r+1}} \dots p_k^{x_k}}{p_1^{x_1} \dots p_r^{x_r} (1 - p_1 - \dots - p_r)^{-(n - x_1 - \dots - x_r)}} \end{aligned}$$

If $X_1, \dots, X_r$ are given, let $x = x_1 + \dots + x_r$ and $p = p_1 + \dots + p_r$

$$f_{X_{r+1}, \dots, X_k | X_1, \dots, X_r}(x_{r+1}, \dots, x_k | x_1, \dots, x_r) = \binom{n-x}{x_{r+1}, \dots, x_k} \frac{p_{r+1}^{x_{r+1}} \dots p_k^{x_k}}{p_1^{x_1} \dots p_r^{x_r} (1-p)^{-(n-x)}}$$

[Textbook Solution]

$$= \binom{n-x}{x_{r+1}, \dots, x_k} \left(\frac{p_{r+1}}{1-p} \right)^{x_{r+1}} \dots \left(\frac{p_k}{1-p} \right)^{x_k}$$

$$\hat{=} \text{Multinomial}(n-x, \left(\frac{p_{r+1}}{1-p}, \dots, \frac{p_k}{1-p} \right)^t)$$

3.5. 확률변수 X 가 기하분포 $\text{Geo}(p)$ ($0 < p < 1$)를 따를 때

$$P(X > k+j | X > k) = P(X > j) \quad (j, k = 0, 1, 2, \dots)$$

임을 밝히라. 이를 기하분포의 무기억성 (memoryless property) 이라고 한다.

$$X \sim \text{Geo}(p), \quad f_X(x) = (1-p)^{x-1} \cdot p, \quad x = 1, 2, \dots$$

$$P(X < k) = \sum_{x=1}^{k-1} (1-p)^{x-1} \cdot p = \sum_{y=0}^{k-2} p(1-p)^y = 1 - (1-p)^{k-1}$$

$$\text{Suppose } S_n = \sum_{x=0}^n ar^x, \quad S_{n+1} = \sum_{x=0}^{n+1} ar^x$$

$$S_{n+1} - S_n = ar^{n+1}$$

$$S_n(r-1) = a(r^{n+1} - 1)$$

$$\therefore S_n = a \cdot \frac{1-r^{n+1}}{1-r}$$

$$\therefore P(X < k) = p \cdot \frac{1 - (1-p)^{k-1}}{1 - (1-p)} = 1 - (1-p)^{k-1}$$

$$\Rightarrow P(X \leq k) = 1 - (1-p)^k$$

$$\therefore P(X > k) = 1 - P(X \leq k) = (1-p)^k$$

$$P(X > j+k, X > k) = P(X > j+k) = (1-p)^{j+k}$$

$$\begin{aligned} \therefore P(X > j+k, X > k) / P(X > k) &= (1-p)^{j+k} / (1-p)^k = (1-p)^j = P(X > j) \\ \therefore P(X > j+k | X > k) &= P(X > j) \end{aligned}$$

[Textbook Solution]

$$P(X > j) = \sum_{q=j+1}^{\infty} q^{x-1} p = q^j, q=1-p$$

$$P(X > k+j | X > k) = P(X > k+j) / P(X > k)$$

3.6. 음이항분포 $\text{Negbin}(r, p)$ 의 확률밀도함수

$$f(x) = \binom{x-1}{r-1} p^r (1-p)^{x-r}, \quad x = r, r+1, \dots$$

에 대하여 다음식 $f(x) \geq f(x-1)$ 을 풀고, 이로써 $f(x)$ 를 최대화하는 x 값을

$$x = [(r-1)/p] + 1$$

임을 밝히라

$$\begin{aligned} f(x) \geq f(x-1) &\Rightarrow \frac{(x-1)!}{(r-1)!(x-r)!} p^r (1-p)^{x-r} \geq \frac{(x-2)!}{(r-1)!(x-r-1)!} p^r (1-p)^{x-r-1} \\ &= \frac{(x-1)!}{(r-1)!(x-r)!} p^r (1-p)^{x-r} \geq \frac{(x-2)!}{(r-1)!(x-r-1)!} p^r (1-p)^{x-r-1} \\ &= (x-1)(1-p) \geq (x-r) \\ &= x(1-p) - (1-p) \geq x-r \\ &= -x \cdot p \geq 1-p-r \\ &= -x \geq (1-r)p^{-1} - 1 \\ &= x \leq (r-1)/p + 1 \quad \therefore x = [(r-1)/p] + 1 \end{aligned}$$

3.7. 확률변수 X 가 음이항분포 $\text{Negbin}(r, p)$ ($0 < p < 1$)을 따를 때 연습문제 11.14)에서 정의된 X 의 왜도와 첨예도를 구하라.

해답: Skewness: $E\left[\left(\frac{X-\mu}{\sigma}\right)^3\right] = C_X(t=0) = C_X(t=0)/\sigma^3$

첨예도: Kurtosis: $E\left[\left(\frac{X-\mu}{\sigma}\right)^4\right] = C_X^{(4)}(t=0) = C_X^{(4)}(t=0)/\sigma^4$

$$z = \frac{X-\mu}{\sigma}$$

$$f_X(x) = \binom{n-1}{x-1} (1-p)^{x-1} p^1$$

$$X \triangleq Z_1 + \dots + Z_r, Z_i \sim \text{Geo}(p)$$

$$f_Z(z) = (1-p)^{z-1} \cdot p, M_Z(t) = [e^{tz}] = \sum_{z=1}^{\infty} e^{tz} \cdot (1-p)^{z-1} \cdot p$$

$$= p \cdot \sum_{z=1}^{\infty} e^{tz} (1-p)^{z-1} = p \left[\sum_{z=0}^{\infty} e^{tz} (1-p)^{z-1} - (1-p)^{-1} \right]$$

$$= p \left[(1-p)^{-1} \sum_{z=0}^{\infty} e^{tz} (1-p)^z - (1-p)^{-1} \right]$$

$$= p \left[(1-p)^{-1} (1-e^{tq})^{-1} - (1-p)^{-1} \right]$$

$$= p \left[(1-p)^{-1} \left\{ \frac{1-1+e^{tq}}{1-e^{tq}} \right\} \right]$$

$$= p \cdot q^{-1} \frac{e^{tq}}{1-e^{tq}} = p \cdot e^t [1-e^{tq}]^{-1}$$

$$\therefore M_X(t) = M_Z(t)^r = [p e^t (1-e^{tq})^{-1}]^r$$

$$C_X(t) = r \log(p) + r t - r \log(1-e^{tq})$$

$$C_X'(t) = r - r \cdot \frac{-e^{tq} \cdot q}{1-e^{tq}} = r + \frac{rq e^t}{1-e^{tq}} = \frac{r - rq e^t + rq e^t}{1-e^{tq}} = r(1-e^{tq})^{-1}$$

$$= r(1-e^{tq})^{-1}$$

$$C_X''(t) = r(q e^t)(1-e^{tq})^{-2}$$

$$C_X'''(t) = r(q e^t)(1-e^{tq})^{-2} + 2r(q e^t)^2(1-e^{tq})^{-3}$$

$$C_X^{(4)}(t) = r(q e^t)(1-e^{tq})^{-2} + 2r(q e^t)^2(1-e^{tq})^{-3} + 4r(q e^t)^3(1-e^{tq})^{-3} + 6r(q e^t)^3(1-e^{tq})^{-4}$$

$$C_X'(t=0) = r/p$$

$$C_X''(t=0) = rq \cdot p^{-2}$$

$$C_X'''(t=0) = rq p^{-2} + 2rq^2 p^{-3} = rq p^{-2} (1 + 2q p^{-1}) = rq p^{-2} \left(\frac{p+q+q}{p} \right) = rq p^{-2} \left(\frac{1+q}{p} \right)$$

$$= rq(1+q) p^{-3}$$

$$C_X^{(4)}(t=0) = rq p^{-2} + 6rq^2 p^{-3} + 6rq^3 p^{-4} = rq p^{-2} (1 + 6q p^{-1} + 6q^2 p^{-2})$$

$$E[X] = r/p, \text{Var}(X) = \sigma^2 = rq p^{-2}$$

$$\text{Skewness} = C_3/\sigma^3 = \frac{rq(1+q)p^{-3}}{(rq p^{-2})^{3/2}} = \frac{1+q}{p^{1/2}} \cdot \frac{1}{p^{1/2}} = (1+q) p^{-1/2} \quad \checkmark$$

$$\text{Kurtosis} = C_4/\sigma^4 = \frac{rq p^{-2} (1 + 6q p^{-1} + 6q^2 p^{-2})}{(rq p^{-2})^2}$$

$$= (1 + 6q p^{-1} + 6q^2 p^{-2}) / (rq p^{-2})$$

$$= r^{-1} (q^{-1} p^2 + 6p + 6q)$$

$$= r^{-1} (q^{-1} p^2 + 6(p+q))$$

$$= r^{-1} (p^2 q^{-1} + 6) \quad \checkmark$$

* 3.8. 서로 독립이고 성공률이 p ($0 < p < 1$) 인 베르누이 시행 $X_1, \dots, X_n, \dots$ 을 관찰할 때 r 번째 성공까지의 시행횟수를 W_r 이라고 하면 $\text{Cov}(W_1, W_r)$ ($r \geq 2$) 을 구하라.

$$W_r \sim \text{Negbin}(r, p)$$

$$f_{W_r}(w) = \binom{r-1}{w-1} (1-p)^{r-1} p^w$$

$$\text{Cov}(W_1, W_r) = E[W_1 W_r] - E[W_1] E[W_r]$$

$$W_r \triangleq W_1 \oplus W_2 - W_1 \oplus \dots \oplus W_r - W_{r-1}$$

$W_1, W_1 - W_{-1}, W_2 - W_{2-1}, \dots, W_r - W_{r-1}$ are mutually exclusive, and have the same distribution of $\text{Geo}(p)$

$$\begin{aligned} E[W_r, W_r] &= E[W_1(W_1 \oplus W_2 - W_1 \oplus \dots \oplus W_r - W_{r-1})] \\ &= E[W_1^2] + E[W_1] \{E[W_2 - W_1] + \dots + E[W_r - W_{r-1}]\} \\ &= E[W_1^2] + (r-1)E[W_1]^2 \end{aligned}$$

$$\begin{aligned} \therefore \text{Cov}(W_1, W_r) &= E[W_1 W_r] - E[W_1] E[W_r] \\ &= \{E[W_1^2] + (r-1)E[W_1]^2\} - E[W_1] \{E[W_1] + E[W_2 - W_1] + \dots + E[W_r - W_{r-1}]\} \\ &= E[W_1^2] - E[W_1]^2 \end{aligned}$$

$$W_1 \sim \text{Geo}(p)$$

$$\begin{aligned} M_{W_1}(w) &= E[e^{tw}] = \sum_{x=1}^{\infty} e^{tx} (1-p)^{x-1} p \\ &= \sum_{x=0}^{\infty} e^{tx} (1-p)^x \cdot (1-p)^{-1} \cdot p \end{aligned}$$

$$= \frac{1}{1-qe^t} \cdot p \cdot (1-p)^{-1} \cdot (1-p)^{-1} \cdot p$$

$$= p(1-p)^{-1} \cdot \left(\frac{1-1+qe^t}{1-qe^t} \right) = p e^t (1-qe^t)^{-1}$$

$$M_X'(t) = p e^t (1-qe^t)^{-1} + p e^t (1-qe^t)^{-2} q e^t$$

$$M_X'(t=0) = p \cdot p^{-1} + p \cdot p^{-2} \cdot q = 1 + p^{-1} q = 1 + \frac{1-p}{p} = \frac{p+1-p}{p} = p^{-1}$$

$$M_X''(t) = p e^t (1-qe^t)^{-1} + p e^t (1-qe^t)^{-2} q e^t + 2p \cdot q \cdot e^t (1-qe^t)^{-2} + 2p e^t (1-qe^t)^{-3} (q e^t)^2$$

$$M_X''(t) = p p^{-1} + p p^{-2} q + 2p q \cdot p^{-2} + 2p \cdot p^{-2} \cdot q^2$$

$$= 1 + q/p + 2q/p + 2q^2/p$$

Too prone to make

$$= p(p^{-1} q + 2q + 2q^2)$$

Calculation mistakes.

$$= p(p^{-1} + 1 - p + 2 - 2p + 2p^2 + 2p + 2)$$

use cgf instead.

$$= p(3p^2 - p + 5)$$

$$C_X(t) = \log M_X(t) = \log p + t - \log(1-qe^t)$$

$$C_X'(t) = 1 + q e^t (1-qe^t)^{-1}, E[X] = C_X'(t=0) = 1 + q/p = 1/p$$

$$C_X''(t) = q e^t (1-qe^t)^{-1} + (q e^t)^2 (1-qe^t)^{-2}$$

$$C_X''(t) = \text{Var}(X) = q/p + q^2/p^2 = q/p(1+q/p) = q/p^2$$

$$\therefore \text{Cov}(W_1, W_r) = \text{Cov}(W_1, W_2) = q/p^2 \quad q=1-p \quad \checkmark$$

* 식 정의하는 것에 신경쓰기

3.9. 포아송분포 Poisson(λ)의 확률밀도함수 $f(x) = e^{-\lambda} \lambda^x / x!$ ($x=0, 1, 2, \dots$)에 대하여
 복응식 $f(x) \geq f(x-1)$ 을 풀고 이로부터 $f(x)$ 가 $x = [\lambda]$ 에서 최대가 되는 것을 밝혀라.

$$f(x) \geq f(x-1) \Rightarrow e^{-\lambda} \lambda^x / x! \geq e^{-\lambda} \lambda^{x-1} / (x-1)! \\ \Rightarrow \lambda \geq x$$

$\therefore f(x)$ is max at $x = \lambda$ ✓

3.10. 확률변수 X 가 포아송분포 Poisson(λ) ($\lambda > 0$)을 따를 때 연습문제 1.14에서 정의된 X 의 왜도와 첨예도를 구하라.

$$f_X(x) = e^{-\lambda} \lambda^x / x!$$

$$M_X(t) = E[e^{tX}] = \sum_{x=0}^{\infty} e^{tx} \cdot e^{-\lambda} \lambda^x / x! \\ = \sum_{x=0}^{\infty} e^{-\lambda} (e^t \lambda)^x / x! = e^{-\lambda} \exp\{e^t \lambda\} = \exp\{\lambda(e^t - 1)\}$$

$$C_X(t) = \log M_X(t) = \lambda e^t - \lambda$$

$$C_X'(t) = \lambda e^t$$

$$C_X''(t) = \lambda e^t$$

$$\vdots$$

$$C_X^{(m)}(t) = \lambda e^t$$

$$\therefore E[X] = \lambda, \text{Var}(X) = \lambda \\ \text{skewness} = \lambda / \lambda^{3/2} = \lambda^{-1/2} \checkmark$$

$$\text{kurtosis} = \lambda / \lambda^2 = \lambda^{-1} \checkmark$$

3.11. 확률변수 $X_1, \dots, X_k$ 가 서로 독립이고 각각이 포아송분포 Poisson(λ_i) ($i=1, 2, \dots, k$)을 따를 때 $X_1 + \dots + X_k = n$ 인 조건에서 $(X_1, \dots, X_k)$ 의 조건부 확률밀도함수를 구하라.

$$f(x_1, \dots, x_k) = f(x_1) \cdots f(x_k) = \left(\frac{e^{-\lambda_1} \lambda_1^{x_1}}{x_1!} \right) \cdots \left(\frac{e^{-\lambda_k} \lambda_k^{x_k}}{x_k!} \right)$$

$$\text{Let } x_1 + \dots + x_{k-1} = x^*$$

$$f(x_1 = x_1, \dots, x_k = x_k, x_1 + \dots + x_k = n)$$

$$= f(x_1 = x_1, x_2 = x_2, \dots, x_{k-1} = x_{k-1}, x_k = n - x^*)$$

$$= \left(\frac{e^{-\lambda_1} \lambda_1^{x_1}}{x_1!} \right) \cdots \left(\frac{e^{-\lambda_{k-1}} \lambda_{k-1}^{x_{k-1}}}{(x_{k-1})!} \right) \cdot \left(\frac{e^{-\lambda_k} \lambda_k^{x^*}}{x^*!} \right)$$

$$M_{X_i}(t) = \exp\{e^t \lambda_i - \lambda_i\}$$

$$M_{X_1+X_2}(t) = \exp\{e^t \lambda_1 - \lambda_1 + e^t \lambda_2 - \lambda_2\} \\ = \exp\{(\lambda_1 + \lambda_2) e^t - (\lambda_1 + \lambda_2)\}$$

$$\therefore X_1 + X_2 \sim \text{Poisson}(\lambda_1 + \lambda_2)$$

$$X_1 + \dots + X_k \sim \text{Poisson}(\lambda_1 + \lambda_2 + \dots + \lambda_k). \text{ Let } \lambda_1 + \dots + \lambda_k = \lambda^*$$

$$P(X_1 + \dots + X_k = n) = e^{-\lambda^*} \lambda^{*n} / n!$$

$$\therefore f(x_1, x_2, \dots, x_k | x_1 + \dots + x_k = n)$$

$$= \frac{n!}{x_1! \cdots x_k!} x^*! \exp\{-\lambda_1 - \lambda_2 - \dots - \lambda_k + \lambda^*\} \left(\frac{\lambda_1}{\lambda^*} \right)^{x_1} \cdots \left(\frac{\lambda_k}{\lambda^*} \right)^{x_k}$$

$$= \left(\frac{n!}{x_1! \cdots x_k!} \right) \left(\frac{\lambda_1}{\sum \lambda_i} \right)^{x_1} \cdots \left(\frac{\lambda_k}{\sum \lambda_i} \right)^{x_k}$$

$$\sim \text{Multinomial}(n, \left(\frac{\lambda_1}{\sum \lambda_i}, \dots, \frac{\lambda_k}{\sum \lambda_i} \right)^t) \checkmark$$

3.12. 발생률이 λ 인 포아송과정 $\{N_t : t \geq 0\}$ 에서 r 번째 현상이 발생할 때까지의 시간을

$$W_r = \min\{t : N_t \geq r\} \quad (r=1, 2, \dots)$$

이라고 할 때 다음에 답하라.

(a) 사건 $\{W_r > t\}$ 을 N_t 에 관한 식으로 표현하고, 다음 등식이 성립하는 이유를 설명하라.

$$\int_t^{\infty} \frac{1}{t! r!} \lambda^r y^{r-1} e^{-\lambda y} dy = \sum_{k=0}^{r-1} \frac{e^{-\lambda t} (\lambda t)^k}{k!}$$

(b) $\text{Cov}(W_1, W_r) \quad (r \geq 2)$ 를 구하라.

(Solution)

(a) N_t denotes the number of events that happened from time 0 to t .
 W_r denotes the minimum time until the r th event occurs.

If $W_r > t$: r th event happened after time t

$\Leftrightarrow N_t \leq t-1$: $(t-1)$ or less events happened until time t .

$$\therefore \{W_r > t\} = \{N_t \leq t-1\}$$

$$W_r \sim \text{Gamma}(r, 1/\lambda) \text{ and } N_t \sim \text{Poisson}(\lambda t) \checkmark$$

$$\therefore P(W_r > t) = \int_t^{\infty} \frac{1}{t! \Gamma(t) (1/\lambda)^t} x^{t-1} \exp\{-x (\frac{1}{\lambda})\} dx$$

$$P(N_t < r-1) = \sum_{x=0}^{r-1} \frac{e^{-\lambda t} (\lambda t)^x}{x!}$$

$\therefore$ 등식이 성립한다.

$$(b) \text{Cov}(W_1, W_r) = \text{Cov}(W_1, W_1 \oplus W_2 \oplus \dots \oplus W_{r-1}) \\ = \text{Cov}(W_1, W_1) = \text{Var}(W_1)$$

$$W_1 \sim \text{Exp}(1/\lambda) = \lambda \exp\{-\lambda w\}$$

$$M_{W_1}(t) = E[e^{tW_1}] = \int_0^{\infty} e^{tw} \cdot \lambda \exp\{-\lambda w\} dw = \int_0^{\infty} \lambda \exp\{(-\lambda + t)w\} dw$$

$$= \lambda \int_0^{\infty} \exp\{(-\lambda + t)w\} dw = \lambda \left[-\frac{1}{\lambda - t} \exp\{(-\lambda + t)t\} \right]_0^{\infty}$$

$$= \lambda (\lambda - t)^{-1} = (1 - t/\lambda)^{-1} \quad (t < \lambda)$$

$$M'(t) = \frac{1}{\lambda} (1 - t/\lambda)^{-2}$$

$$E[W] = M'(t=0) = \lambda^{-1}$$

$$C_W(t) = \log M_W(t) = -\log(1 - \frac{t}{\lambda})$$

$$C'(t) = \frac{1}{\lambda} (1 - \frac{t}{\lambda})^{-1}$$

$$C''(t) = \frac{1}{\lambda^2} (1 - \frac{t}{\lambda})^{-2}$$

$$\therefore \text{Var}(W) = \lambda^{-2}$$

$$\therefore \text{Cov}(W_1, W_r) = \lambda^{-2} \checkmark$$

3.13. 확률변수 X 가 지수분포 $\text{Exp}(1/\lambda)$ ($\lambda > 0$) 일 때

$$P(X > x+y | X > x) = P(X > y) \quad (x > 0, y > 0)$$

$$f_X(x) = \lambda \exp(-\lambda x) \quad (0 < x < \infty)$$

$$F_X(x) = \int_0^x \lambda e^{-\lambda x} dx = [-e^{-\lambda x}]_0^x = 1 - e^{-\lambda x}$$

$$P(X > x) = 1 - P(X \leq x) = 1 - (1 - e^{-\lambda x}) = e^{-\lambda x}$$

$$P(X > x+y, X > x) = P(X > x+y) = e^{-\lambda(x+y)}$$

$$P(X > y) = e^{-\lambda y}$$

$$\therefore P(X > x+y | X > x) = e^{-\lambda(x+y)} / e^{-\lambda x} = e^{-\lambda y} = P(X > y)$$

$$\therefore P(X > x+y | X > y) = P(X > x)$$

3.14. 감마분포 $\text{Gamma}(a, \beta)$ 의 확률밀도함수

$$f(x) = \frac{1}{\Gamma(a)\beta^a} x^{a-1} e^{-x/\beta} I_{(0, \infty)}(x)$$

에 대하여 $d > 1$ 일 경우에 $f(x)$ 를 최대로 하는 x 값을 $x = (a-1)\beta$ 임을 밝히라.

$$\begin{aligned} f'(x) &= \frac{1}{\Gamma(a-1)\beta^a} x^{a-2} e^{-x/\beta} - \frac{1}{\Gamma(a)\beta^{a+1}} x^{a-1} e^{-x/\beta} \\ &= \frac{1}{\Gamma(a-1)\beta^a} x^{a-2} e^{-x/\beta} \left(1 - \frac{1}{(a-1)\beta} x\right) \quad \therefore x = (a-1)\beta, f'(x) = 0 \end{aligned}$$

You can calculate $f''(x)$ and show that $f''(x) = (a-1)\beta < 0$

But you can also think about the general shape of $\text{Gamma}(a, \beta)$ distributions.

[Textbook Solution]

$$(\log f(x))' = (a-1)/x - 1/\beta > 0$$

$$\Leftrightarrow (a-1)/x > 1/\beta$$

$$\Leftrightarrow x < (a-1)\beta$$

3.15. 확률변수 X 가 감마분포 $\text{Gamma}(a, \beta)$ 를 따를 때 연습문제 (1.14)에서 정의된 X 의 왜도와 첨예도를 구하라.

$$X \sim \text{Gamma}(a, \beta) \quad f_X(x) = \frac{1}{\Gamma(a)\beta^a} x^{a-1} e^{-x/\beta} \quad (x > 0)$$

$$\begin{aligned} M_X(t) &= E[e^{tX}] = \int_0^\infty e^{tx} \frac{1}{\Gamma(a)\beta^a} x^{a-1} e^{-x/\beta} dx \\ &= \int_0^\infty \frac{1}{\Gamma(a)\beta^a} x^{a-1} e^{-(1/\beta - t)x} dx = \int_0^\infty \frac{1}{\Gamma(a)\beta^a} x^{a-1} \exp\left\{-\left(\frac{1-\beta t}{\beta}\right)x\right\} dx \\ &= \int_0^\infty \frac{1}{\Gamma(a)\beta^a} x^{a-1} \exp\left\{-x/\left(\frac{\beta}{1-\beta t}\right)\right\} dx \end{aligned}$$

$$\text{let } \beta^* = \beta/(1-\beta t)$$

$$= \beta^{-a} (\beta^*)^a \int_0^\infty \frac{1}{\Gamma(a)(\beta^*)^a} x^a \exp\{-x/\beta^*\} dx$$

$$= \beta^{-a} \beta^a (1-\beta t)^{-a} = (1-\beta t)^{-a} \quad \text{: memorize!!}$$

$$C_X(t) = \log M_X(t) = -a \log(1-\beta t)$$

$$C_X'(t) = a\beta(1-\beta t)^{-1} \quad C_X^{(3)}(t) = 2a\beta^3(1-\beta t)^{-3} \quad \text{skewness} = 2a\beta^3/(a\beta^3)^{3/2} = 2a^{-1/2} \quad \checkmark$$

$$C_X''(t) = a\beta^2(1-\beta t)^{-2} \quad C_X^{(4)}(t) = 6a\beta^4(1-\beta t)^{-4} \quad \text{Kurtosis} = 6a\beta^4/(a\beta^3)^2 = 6a^{-1} \quad \checkmark$$

3.16. 확률변수 X 의 적률생성함수가 존재하고

$$E[X^{2k}] = (2k)! / (2^k k!), \quad E[X^{2k+1}] = 0, \quad k=1, 2, \dots$$

일 때 X 의 적률생성함수와 확률밀도함수를 구하라.

$$\begin{aligned} M_X(t) &= m_0 + \frac{m_1}{1!} t + \frac{m_2}{2!} t^2 + \frac{m_3}{3!} t^3 + \dots + \frac{m_k}{k!} t^k \\ &= 1 + \sum_{k=1}^\infty \frac{(2k)!}{2^k k! (2k)!} t^{2k} = \sum_{k=0}^\infty \frac{1}{k!} \left(\frac{t^2}{2}\right)^k = \exp\left\{\frac{t^2}{2}\right\} \end{aligned}$$

$$\text{mgf of } N(\mu, \sigma^2) : \exp\left\{\mu t + \frac{1}{2}\sigma^2 t^2\right\} \quad \therefore \exp\left\{\frac{t^2}{2}\right\} : \text{mgf of } N(0, 1)$$

$$\text{pdf of } N(0, 1) : \frac{1}{\sqrt{2\pi}} \exp\left\{-\frac{1}{2}x^2\right\}$$

3.17. 확률변수 X 가 정규분포 $N(\mu, \sigma^2)$ ($\sigma > 0$) 을 따를 때 연습문제 (1.14)에서 정의된 X 의 왜도와 첨예도를 구하라.

$$\begin{aligned} M_X(t) &= \exp\left\{\mu t + \frac{1}{2}\sigma^2 t^2\right\} \\ C_X(t) &= \log C_X(t) = \mu t + \frac{1}{2}\sigma^2 t^2 \\ C_X'(t) &= \mu + \sigma^2 t \\ C_X''(t) &= \sigma^2 \\ C_X^{(3)}(t) &= 0 \quad \therefore \text{skewness} = C_3/\sigma^3 = 0 \\ C_X^{(4)}(t) &= 0 \quad \text{kurtosis} = C_4/\sigma^4 = 0 \end{aligned}$$

3.18. 양수의 값을 취하는 확률변수 X 에 대하여 $\log X \sim N(\mu, \sigma^2)$ 일 때 X 의 분포를 로그정규분포 (log normal distribution)라고 하며 기호로는 $X \sim \text{Lognormal}$ 이라고 나타낸다. 이러한 로그분포에 관한 다음 성질을 밝히라.

- (a) $X \sim \text{Lognormal}(\mu, \sigma^2)$ 이면 $E[X^K] = \exp\left\{K\mu + \frac{1}{2}K^2\sigma^2\right\}$, $K=1, 2, \dots$
 (b) $X \sim \text{Lognormal}(0, 1)$ 일 때 X 의 적률생성함수는 존재하지 않는다.

$$\text{[참고:]} \int_{x > \sqrt{12}/t} \exp\{te^x\} \frac{1}{\sqrt{2\pi}} e^{-x^2/2} dx = +\infty, \quad \forall t > 0$$

$$\begin{aligned} Y &= \log X \quad E[Y] = \mu, \text{Var}(Y) = \sigma^2 \\ M_Y(t) &= E[e^{tY}] = E[X^t] \\ &= \exp\left\{\mu t + \frac{1}{2}t^2\sigma^2\right\} \end{aligned}$$

$$\begin{aligned} \therefore (a) &\Rightarrow \exp\left\{\mu t + \frac{1}{2}t^2\sigma^2\right\} = M_Y(t) = E[e^{tY}] = E[(e^{\log X})^t] = E[X^t] \\ &\Rightarrow E[X^t] = \exp\left\{\mu t + \frac{1}{2}t^2\sigma^2\right\} \end{aligned}$$

$$(b) M_X(t) = E[e^{tX}] = E[(e^X)^t] = E[\exp\{e^Y\}]$$

$$= \int_{-\infty}^{+\infty} \exp\{te^y\} \frac{1}{\sqrt{2\pi}} \exp\left\{-\frac{1}{2}y^2\right\} dy$$

$$= \int_{-\infty}^{+\infty} \exp\{te^y\} \frac{1}{\sqrt{2\pi}} \exp\left\{-\frac{1}{2}y^2\right\} dy + \int_{\sqrt{12}/t}^{+\infty} \exp\{te^y\} \frac{1}{\sqrt{2\pi}} \exp\left\{-\frac{1}{2}y^2\right\} dy$$

$$\begin{aligned} \exp\{te^y\} &\leq \exp\{te^{\sqrt{12}/t}\} \quad \forall y \in (-\infty, \sqrt{12}/t) \\ \therefore \int_{-\infty}^{\sqrt{12}/t} \exp\{te^y\} \frac{1}{\sqrt{2\pi}} \exp\left\{-\frac{1}{2}y^2\right\} dy &\leq \int_{-\infty}^{\sqrt{12}/t} \exp\{te^{\sqrt{12}/t}\} \frac{1}{\sqrt{2\pi}} \exp\left\{-\frac{1}{2}y^2\right\} dy \\ &= M \in \mathbb{R} \end{aligned}$$

$$\begin{aligned} \therefore &= \int_{\sqrt{12/t}}^{+\infty} \exp\{te^y\} \frac{1}{\sqrt{2\pi}} \exp\{-\frac{1}{2}y^2\} dy + \boxed{\int_{-\infty}^{\sqrt{12/t}} \exp\{te^y\} \frac{1}{\sqrt{2\pi}} \exp\{-\frac{1}{2}y^2\} dy} \\ &\geq +\infty + M \geq +\infty \end{aligned}$$

$\therefore$ MGF of $X \sim \text{Lognormal}(0,1)$ does not exist.

[Textbook Solution]

(a) $X \stackrel{d}{=} \text{Exp}(Z)$, $Z \sim N(\mu, \sigma^2)$

$$E[X^k] = E[\exp(kZ)], Z \sim N(\mu, \sigma^2)$$

$$= \text{mgf}_{N(\mu, \sigma^2)}(k)$$

$$= \exp\{k\mu + k^2\sigma^2/2\}$$

(b) $X \stackrel{d}{=} \text{Exp}(Z)$, $Z \sim N(0,1)$

$$\forall t > 0, \int_{-\infty}^{+\infty} e^{tx} p d f_X(x) dx = \int_{-\infty}^{+\infty} \exp\{te^x\} \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}x^2} dx$$

$$\geq \int_{z \geq \sqrt{12/t}} \exp\{te^z\} \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}z^2} dz$$

$$\geq \int_{z \geq \sqrt{12/t}} \exp\{tz\} dz / \sqrt{2\pi}$$

$$= +\infty$$

Ask
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